# Problem 1

**(a)** *Express $P(X_i = 1)$ as a function of $a$ and $\pi$.*

We have

$$\begin{aligned} P(X_i = 1) &= P(X_i = 1, \text{Group} = A) + P(X_i = 1, \text{Group} = B) \\ &= P(X_i = 1 \mid \text{Group} = A) \cdot P(\text{Group} = A) + P(X_i = 1 \mid \text{Group} = B) \cdot P(\text{Group} = B) \\ &= a \cdot P(\text{Group} = A) + (1-a) \cdot P(\text{Group} = B) \\ &= a \cdot \pi + (1-a) \cdot (1-\pi) \ . \end{aligned}$$

Naturally, we have that

$$\begin{aligned} P(X_i = 0) &= P(X_i = 0, \text{Group} = A) + P(X_i = 0, \text{Group} = B) \\ &= P(X_i = 0 \mid \text{Group} = A) \cdot P(\text{Group} = A) + P(X_i = 0 \mid \text{Group} = B) \cdot P(\text{Group} = B) \\ &= (1-a) \cdot P(\text{Group} = A) + a \cdot P(\text{Group} = B) \\ &= (1-a) \cdot \pi + a \cdot (1-\pi) \ . \end{aligned}$$

*(Optional)* Because the random variable $X_i$ can only have 2 potential values, we can now easily check whether our expressions are correct as $P(X_i = 1) + P(X_i = 0) = 1$. We have

$$\begin{aligned} P(X_i = 1) + P(X_i = 0) &= a \cdot \pi + (1-a) \cdot (1-\pi) + (1-a) \cdot \pi + a \cdot (1-\pi) \\ &= a \cdot \pi + (1-\pi) - a + \pi \cdot a + \pi - \pi \cdot a + a - \pi \cdot a \\ &= a \cdot \pi + 1 - \pi + \pi - \pi \cdot a \\ &= 1 \ . \end{aligned}$$

**(b)** *Give expressions for the joint probability of the observed data $X_1, X_2, \ldots, X_n$, and the corresponding likelihood function and loglikelihood function.*

We aim to express the computation for

$$P(X_1, X_2, \ldots, X_n) \ .$$

Because we know that the $n$ samples are drawn with the i.i.d. assumption, it holds that

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P(X_i) \ .$$

Hence,

$$
\begin{aligned}
P(X_1, X_2, \dots, X_n) &= \prod_{i=1}^{n} P(X_i) \\
&= \prod_{i=1}^{n} \begin{cases} P(X_i = 1) & \text{if } X_i = 1 \\ P(X_i = 0) & \text{if } X_i = 0 \end{cases} \\
&= P(X_i = 1)^m \cdot P(X_i = 0)^{n-m} \\
&= (a \cdot \pi + (1-a) \cdot (1-\pi))^m \cdot ((1-a) \cdot \pi + a \cdot (1-\pi))^{n-m} \qquad (1)
\end{aligned}
$$

which is the likelihood function of the observed samples.

Let us now derive the experession for the log likelihood of the data:

$$
\begin{aligned}
\log P(X_1, X_2, \dots, X_n) &= \log \left[ (a \cdot \pi + (1-a) \cdot (1-\pi))^m \cdot ((1-a) \cdot \pi + a \cdot (1-\pi))^{n-m} \right] \\
&= m \cdot \log \left[ a \cdot \pi + (1-a) \cdot (1-\pi) \right] + (n-m) \cdot \log \left[ (1-a) \cdot \pi + a \cdot (1-\pi) \right] \qquad (2)
\end{aligned}
$$

**(c)** *Derive an expression for the maximum likelihood estimator* $\pi_{ML}$ *of* $\pi$.

We now aim to estimate $\pi$. To this end, we replace $\pi$ with $\pi_{ML}$ in Equation 2 and derive the log likelihood w.r.t. $\pi_{ML}$. We have

$$
\begin{aligned}
&\frac{\partial}{\partial \pi_{ML}} \log P(X_1, X_2, \dots, X_n) \\
=&\frac{\partial}{\partial \pi_{ML}} m \cdot \log \left[ a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML}) \right] + (n-m) \cdot \log \left[ (1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML}) \right] \\
=&m \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} \cdot (a - (1-a)) + (n-m) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \cdot ((1-a) - a) \,.
\end{aligned}
$$

We can now find the critical points by setting the gradient to zero:

$$
\begin{aligned}
\frac{\partial}{\partial \pi_{ML}} \log P(X_1, X_2, \dots, X_n) &\overset{!}{=} 0 \\
m \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} \cdot (a - (1-a)) + (n-m) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \cdot ((1-a) - a) &= 0 \qquad (3)
\end{aligned}
$$

where we observe that the the equation holds whenever $(1-a) = a$, because then both additive terms are zero. It is easy to see that $a = 0.5$ is the only real valued solution for this condition. However, as we are interested in the maximum likelihood estimator of $\pi_{ML}$, setting $a = 0.5$ prohibits us from finding the maximum value.

Explain why setting $a = 0.5$ is problematic in more details...

Therefore, we have to consider the case where $a \neq 0.5$.

$$
\begin{aligned}
m \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} \cdot (a-(1-a)) &= -(n-m) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \cdot ((1-a)-a) \\
m \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} \cdot (a-(1-a)) &= (n-m) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \cdot (a-(1-a)) \\
m \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} &= (n-m) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \\
m \cdot \frac{1}{(2a-1) \cdot \pi_{ML} + 1 - a} &= (n-m) \cdot \frac{1}{(1-2a) \cdot \pi_{ML} + a} \\
m \cdot \frac{1}{(2a-1) \cdot \pi_{ML} + 1 - a} &= (n-m) \cdot \frac{1}{a - (2a-1) \cdot \pi_{ML}} \\
m \cdot (a - (2a-1) \cdot \pi_{ML}) &= (n-m) \cdot ((2a-1) \cdot \pi_{ML} + 1 - a) \\
am - (2a-1) \cdot m \cdot \pi_{ML} &= (n-m) \cdot (2a-1) \cdot \pi_{ML} + (n-m) \cdot (1-a) \\
am - (n-m) \cdot (1-a) &= (n-m) \cdot (2a-1) \cdot \pi_{ML} + (2a-1) \cdot m \cdot \pi_{ML} \\
am - (n-m) \cdot (1-a) &= \pi_{ML} \cdot (2a-1)((n-m)+m) \\
am - (n-m) + a \cdot n - am &= \pi_{ML} \cdot (2a-1)(n) \\
n \cdot (a-1) + m &= \pi_{ML} \cdot (2a-1)(n) \\
\frac{n \cdot (a-1) + m}{(2a-1)(n)} &= \pi_{ML} \\
\frac{n \cdot (a-1) + m}{n} \cdot \frac{1}{2a-1} &= \pi_{ML} \\
\left(a - 1 + \frac{m}{n}\right) \cdot \frac{1}{2a-1} &= \pi_{ML} \qquad (4)
\end{aligned}
$$

We verify that this is indeed a maximum by plugging the result from Equation 4 into the second derivative:

$$
\begin{aligned}
&\frac{\partial^2}{\partial {\pi_{ML}}^2} \log P(X_1, X_2, \ldots, X_n) \\
=&\frac{\partial}{\partial \pi_{ML}} m \cdot \frac{1}{a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML})} \cdot (a-(1-a)) + (n-m) \cdot \frac{1}{(1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML})} \cdot ((1-a)-a) \\
=&m \cdot (a-(1-a)) \cdot \frac{1}{(a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML}))^2} \cdot (a-(1-a)) \\
&+ (n-m) \cdot ((1-a)-a) \cdot \frac{1}{((1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML}))^2} \cdot ((1-a)-a) \\
=&2m \cdot (a-(1-a)) \cdot \frac{1}{(a \cdot \pi_{ML} + (1-a) \cdot (1-\pi_{ML}))^2} + 2(n-m) \cdot ((1-a)-a) \cdot \frac{1}{((1-a) \cdot \pi_{ML} + a \cdot (1-\pi_{ML}))^2}
\end{aligned}
$$

This will be loooong I think...

**(d)** *Can you think of an easier way to arrive at the same estimator as the maximum likelihood estimator?*

We observe that $X_i$ follows a Bernoulli distribution. We now first show that this holds.

Recall that the definition of a Bernoulli distribution is given by

$$
\begin{aligned}
P(X = 1) &= p \\
P(X = 0) &= 1 - p \; .
\end{aligned}
$$

Therefore, we now aim to show that $P(X_i)$ follows a Bernoulli distribution. From (a), we have that We have

$$
P(X_i = 1) = a \cdot \pi + (1 - a) \cdot (1 - \pi)
$$

and

$$
P(X_i = 0) = (1 - a) \cdot \pi + a \cdot (1 - \pi) \; .
$$

We multiply out the expression to get

$$
\begin{aligned}
P(X_i = 1) &= a \cdot \pi + 1 - \pi - a \cdot (1 - \pi) \\
&= a \cdot \pi + 1 - \pi - a + a \cdot \pi \\
&= 2a \cdot \pi + 1 - \pi - a \\
&= 1 + 2a \cdot \pi - \pi - a \qquad (5) \\
P(X_i = 0) &= (1 - a) \cdot \pi + a \cdot (1 - \pi) \\
&= \pi - a \cdot \pi + a - a \cdot \pi \\
&= -2a \cdot \pi + \pi + a \; . \qquad (6)
\end{aligned}
$$

Therefore, with $p = 1 + 2a \cdot \pi - \pi - a$ (from Equation 5), we have

$$
\begin{aligned}
P(X = 0) &= 1 - p \\
&= 1 - P(X = 1) \\
&= 1 - (1 + 2a \cdot \pi - \pi - a) \\
&= 1 - 1 - 2a \cdot \pi + \pi + a \\
&= -2a \cdot \pi + \pi + a \qquad (7)
\end{aligned}
$$

where Equation 7 matches the expression in Equation 6. Therefore, we have shown that $X_i$ follows a Bernoulli distribution with parameter $p = 1 + 2a \cdot \pi - \pi - a$.

Following from this result, because the random variable $m$ is the sum of $n$ i.i.d. Bernoulli random variables, we know that $m$ follows a Binomial distribution with parameters $n$ and $p$. Hence, the likelihood of observing $m$ ones in $n$ samples is given by the probability density function of the Binomial distribution:

$$
\Pr(X = m) = \binom{n}{m} p^m (1 - p)^{n-m} \; . \qquad (8)
$$

While Equation 8 differs from the likelihood function we derived in Equation 1, we can see that the $\binom{n}{m}$ term cancels out when deriving the log likelihood; First, the log likelihood transforms the multiplication into a

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sum with a log term, and then when deriving w.r.t. $\pi_{ML}$, the log $\binom{n}{m}$ becomes zero and therefore disappears. Hence, we end up with the same expression for the log likelihood.

As the Binomial distribution is well-known, we know that the maximum likelihood estimator of $p$ of the Binomial distribution is given by $p_{ML} = \frac{m}{n}$, which is the proportion of successes, i.e., the number of $X_i = 1$ in $n$ samples.[1]

Finally, to derive the maximum likelihood estimator of $\pi_{ML}$, we equate $p_{ML}$ and $P(X_i = 1)$ (as $P(X_i = 1) = p$), and then solve for $\pi_{ML}$:

$$\begin{aligned}
P(X_i = 1) &= p_{ML} \\
1 + 2a \cdot \pi_{ML} - \pi_{ML} - a &= \frac{m}{n} \\
2a \cdot \pi_{ML} - \pi_{ML} &= \frac{m}{n} + a - 1 \\
\pi_{ML} \cdot (2a - 1) &= \frac{m}{n} + a - 1 \\
\pi_{ML} &= \frac{\frac{m}{n} + a - 1}{(2a - 1)} \; . \qquad (9)
\end{aligned}$$

We clearly observe that Equation 9 matches Equation 4, hence, we found the same maximum likelihood estimator of $\pi_{ML}$ as in (c).

**(e)** *Show that $\pi_{ML}$ is an unbiased estimator of $\pi$.*

We aim to show that $\mathbb{E}[\pi_{ML}] = \pi$. To this end, we will make use of the knowledge introduced in (d), namely that $X_i$ follows a Bernoulli distribution.

We have

$$\begin{aligned}
\mathbb{E}\left[\pi_{ML}\right] &= \mathbb{E}\left[\frac{\frac{m}{n} + a - 1}{(2a - 1)}\right] \\
&= \mathbb{E}\left[\frac{\frac{\sum_{i=1}^{n} X_i}{n} + a - 1}{(2a - 1)}\right] \\
&= \frac{\sum_{i=1}^{n} \mathbb{E}[X_i] + a \cdot n - 1 \cdot n}{(2a - 1) \cdot n} \; . \qquad (10)
\end{aligned}$$

We now compute the expectation of $X_i$:

$$\begin{aligned}
\mathbb{E}[X_i] &= P(X_i = 1) \cdot 1 + P(X_i = 0) \cdot 0 \\
&= P(X_i = 1) \\
&= a \cdot \pi + (1 - a) \cdot (1 - \pi) \\
&= (1 - a) + (2a - 1) \cdot \pi \; . \qquad (11)
\end{aligned}$$

[1] See `https://en.wikipedia.org/wiki/Binomial_distribution#Estimation_of_parameters`.

Finally, we can substitute Equation 11 into Equation 10 to get

$$
\begin{aligned}
\mathbb{E}\left[\pi_{ML}\right] &= \frac{\sum_{i=1}^{n} \mathbb{E}[X_i] + a \cdot n - 1 \cdot n}{(2a-1) \cdot n} \\
&= \frac{\sum_{i=1}^{n} \left((1-a) + (2a-1) \cdot \pi\right) + a \cdot n - 1 \cdot n}{(2a-1) \cdot n} \\
&= \frac{n \cdot \left((1-a) + (2a-1) \cdot \pi\right) + a \cdot n - 1 \cdot n}{(2a-1) \cdot n} \\
&= \frac{\left((1-a) + (2a-1) \cdot \pi\right) + a - 1}{(2a-1)} \\
&= \frac{-(a-1) + (2a-1) \cdot \pi + (a-1)}{(2a-1)} \\
&= \frac{(2a-1) \cdot \pi}{(2a-1)} \\
&= \pi \ . \qquad (12)
\end{aligned}
$$

We have shown that the expected value of $\pi_{ML}$ is equal to the true value of $\pi$ (Equation 12). Therefore, we can conclude that our estimator for $\pi_{ML}$ is unbiased.

**(f)** *Derive an expression for the variance of $\pi_{ML}$. Analyse its dependence on $a$.*

We have

$$
\begin{aligned}
\mathbb{V}[\pi_{ML}] &= \mathbb{V}\left[\frac{\frac{m}{n} + a - 1}{(2a-1)}\right] \\
&= \mathbb{V}\left[\frac{m}{n} + a - 1\right] \cdot \frac{1}{(2a-1)^2} \\
&= \mathbb{V}\left[\frac{1}{n}\left(m + a \cdot n - 1 \cdot n\right)\right] \cdot \frac{1}{(2a-1)^2} \\
&= \mathbb{V}\left[m + a \cdot n - 1 \cdot n\right] \cdot \frac{1}{(2a-1)^2 \cdot n^2} \\
&= \mathbb{V}\left[\sum_{i=1}^{n} X_i + a \cdot n - 1 \cdot n\right] \cdot \frac{1}{(2a-1)^2 \cdot n^2} \\
&= \mathbb{V}\left[\sum_{i=1}^{n} X_i\right] \cdot \frac{1}{(2a-1)^2 \cdot n^2} \\
&= \sum_{i=1}^{n} \mathbb{V}\left[X_i\right] \cdot \frac{1}{(2a-1)^2 \cdot n^2} \ . \qquad (13)
\end{aligned}
$$

We now compute the variance of $X_i$. We have:

$$
\begin{aligned}
\mathbb{E}[X_i^2] &= P(X_i = 1) \cdot 1^2 + P(X_i = 0) \cdot 0^2 \\
&= P(X_i = 1) \\
&= (1-a) + (2a-1) \cdot \pi \ , \qquad (14)
\end{aligned}
$$

and

$$
\begin{aligned}
\mathbb{E}[X_i]^2 &= ((1-a) + (2a-1)\cdot\pi)^2 \\
&= (1-a)^2 + (2a-1)^2 \cdot \pi^2 + 2\cdot(1-a)\cdot(2a-1)\cdot\pi \\
&= 1^2 + a^2 - 2a + 4a^2\pi^2 + \pi^2 - 4a\cdot\pi^2 + (2-2a)\cdot(2a-1)\cdot\pi \\
&= 1 + a^2 - 2a + 4a^2\pi^2 + \pi^2 - 4a\cdot\pi^2 + (4a - 2 - 4a^2 + 2a)\cdot\pi \\
&= 1 + \pi\left(6a - 4a^2 - 2\right) + \pi^2(4a^2 - 4a + 1) - 2a + a^2 \; . \qquad (15)
\end{aligned}
$$

Therefore, the variance of $X_i$ is given by

$$
\begin{aligned}
V(X_i) &= \mathbb{E}\left[X_i^2\right] - E[X_i]^2 \\
&= \pi(2a-1) + 1 - a - \left(1 + \pi\left(6a - 4a^2 - 2\right) + \pi^2(4a^2 - 4a + 1) - 2a + a^2\right) \\
&= \pi(2a-1) + 1 - a - 1 - \pi\left(6a - 4a^2 - 2\right) - \pi^2(4a^2 - 4a + 1) + 2a - a^2 \\
&= \pi(2a-1) - \pi\left(6a - 4a^2 - 2\right) - \pi^2(4a^2 - 4a + 1) + a - a^2 \\
&= \pi\left(2a - 1 - (6a - 4a^2 - 2)\right) - \pi^2(4a^2 - 4a + 1) + a - a^2 \\
&= \pi\left(2a - 1 - 6a + 4a^2 + 2\right) - \pi^2(4a^2 - 4a + 1) + a - a^2 \\
&= \pi\left(1 - 4a + 4a^2\right) - \pi^2(1 - 4a + 4a^2) + a - a^2 \\
&= \left(1 - 4a + 4a^2\right)\cdot(\pi - \pi^2) + a - a^2 \; . \qquad (16)
\end{aligned}
$$

Finally, to compute the variance of the estimator of $\pi_{ML}$, we plug Equation 16 into Equation 13 and get

$$
\begin{aligned}
\mathbb{V}[\pi_{ML}] &= \sum_{i=1}^{n} \mathbb{V}\left[X_i\right]\cdot\frac{1}{(2a-1)^2\cdot n^2} \\
&= \sum_{i=1}^{n}\left(\left(1 - 4a + 4a^2\right)\cdot(\pi - \pi^2) + a - a^2\right)\cdot\frac{1}{(2a-1)^2\cdot n^2} \\
&= n\cdot\left(\left(1 - 4a + 4a^2\right)\cdot(\pi - \pi^2) + a - a^2\right)\cdot\frac{1}{(2a-1)^2\cdot n^2} \\
&= \left(\left(1 - 4a + 4a^2\right)\cdot(\pi - \pi^2) + a - a^2\right)\cdot\frac{1}{(2a-1)^2\cdot n} \\
&= \left((2a-1)^2\cdot(\pi - \pi^2) + a - a^2\right)\cdot\frac{1}{(2a-1)^2\cdot n} \\
&= \frac{(2a-1)^2\cdot(\pi - \pi^2)}{(2a-1)^2\cdot n} + \frac{a - a^2}{(2a-1)^2\cdot n} \\
&= \frac{\pi - \pi^2}{n} + \frac{a - a^2}{(2a-1)^2\cdot n} \; .
\end{aligned}
$$

We correctly observe that the variance is undefined for $a = 0.5$. Furthermore, we observe that the variance decreases as $n$ increases, which is to be expected as we get more evidence with more data. Lastly, we observe that whenever $a$ approaches either 0 or 1, the variance decreases. The same phenomenon is observed for the true value of $\pi$. Figure 1 shows the log-scaled variance of $\pi_{ML}$ as a function of $a$ and $\pi$, which clearly displays the behavior previously described.

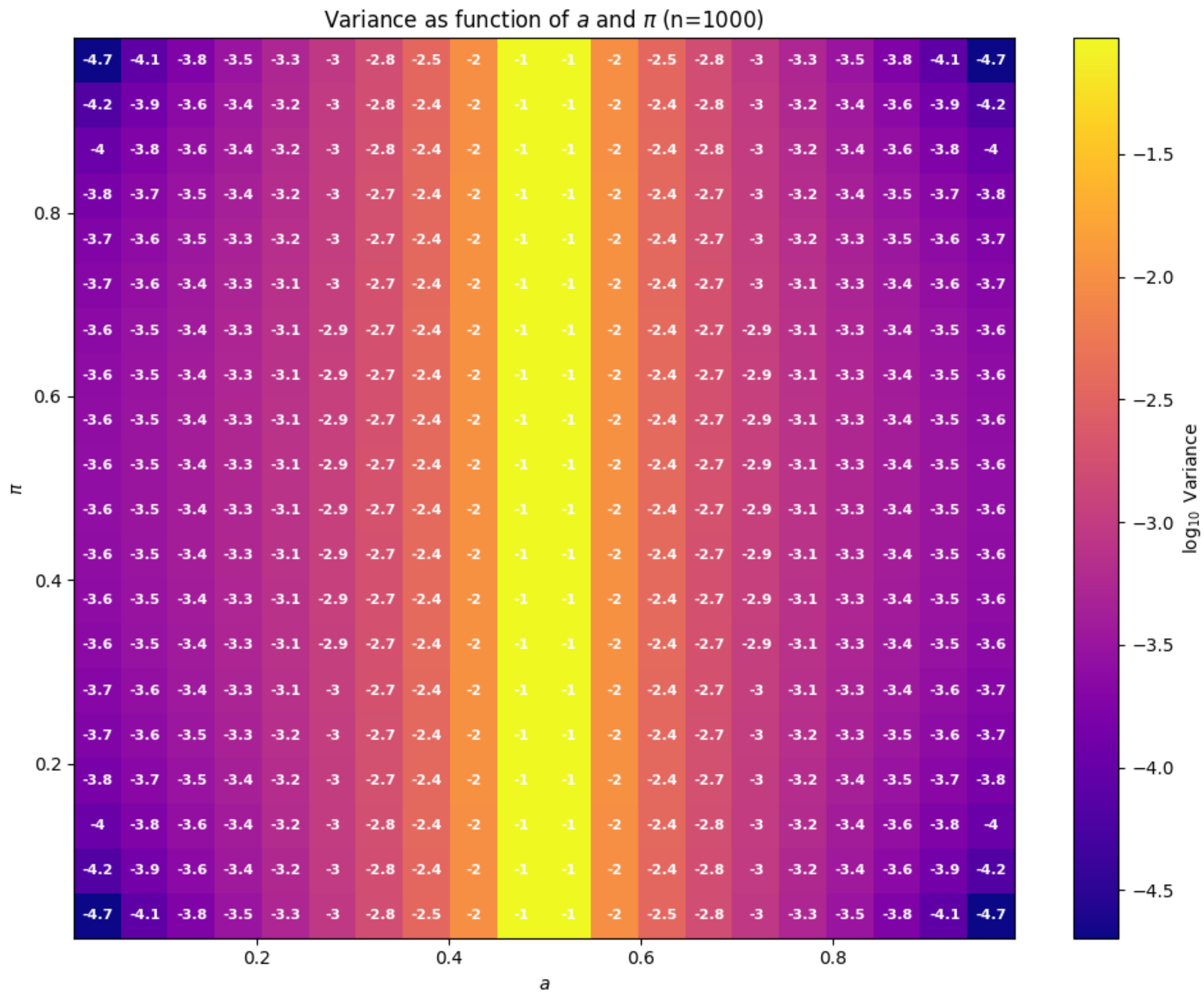


Figure 1: Variance of $\pi_{ML}$ as function of $a$ and $\pi$ for different values of $n$. Here, the values are log-scaled for better display. Note that we restrict $a \in [0.01, 0.99]$ and $\pi \in [0.01, 0.99]$ for the plot. Hence, very low values are expected whenever $a$ or $\pi$ exceeds these bounds.

**(g)** *Give an expression for the bias of this naive estimator. Is it asymptotically unbiased?*

We aim to recover the value of $\pi$ from direct questioning. However, persons belonging to the sensitive group will lie with probability $\ell$, whereas persons belonging to the non-sensitive group will answer truthfully and therefore lie with probability 0.

Naivily, we could estimate the value of $\pi$ as the fraction of persons answering yes to the question:

$$\pi_{\text{naive}} = \frac{\sum_{i=1}^{n} X_i}{n} \ .$$

We can model the whether a person belongs to the sensitive group as a hidden variable $S_i \in \{0, 1\}$ with $S_i = 1$ describing that the person belongs to the sensitive group, and $S_i = 0$ describes that a person belongs to the non-sensitive group.

Naturally, we have that

$$\begin{aligned}P(S_i = 1) &= \pi \\ P(S_i = 0) &= 1 - \pi\end{aligned}$$

which is distributed according to a Bernoulli distribution with parameter $p = \pi$.

We can now express the probability of a person answering yes or no to the question by marginalizing over the hidden variable $S_i$:

$$\begin{aligned}P(X_i = 1) &= P(X_i = 1, S_i = 1) + P(X_i = 1, S_i = 0) \\ &= \pi \cdot (1 - \ell) + 0 \\ P(X_i = 0) &= P(X_i = 0, S_i = 1) + P(X_i = 0, S_i = 0) \\ &= \pi \cdot \ell + (1 - \pi)\ .\end{aligned}$$

We verify that the probabilities are correct by adding both $P(X_i = 1)$ and $P(X_i = 0)$ and getting 1:

$$\begin{aligned}P(X_i = 1) + P(X_i = 0) &= \pi \cdot (1 - \ell) + \pi \cdot \ell + (1 - \pi) \\ &= \pi(1 - \ell + \ell) + 1 - \pi \\ &= \pi(1) + 1 - \pi \\ &= 1\ .\end{aligned}$$

We now compute the expeced value of the naive estimator of $\pi$ to check whether the estimator is unbiased. We have

$$\begin{aligned}\mathbb{E}[\pi_{\text{naive}}] &= \mathbb{E}\left[\frac{\sum_{i=1}^n X_i}{n}\right] \\ &= \frac{1}{n} \cdot \sum_{i=1}^n \mathbb{E}[X_i] \\ &= \frac{1}{n} \cdot \sum_{i=1}^n \left(P(X_i = 1) \cdot 1 + P(X_i = 0) \cdot 0\right) \\ &= \frac{1}{n} \cdot \sum_{i=1}^n \left(P(X_i = 1)\right) \\ &= \frac{1}{n} \cdot \sum_{i=1}^n \left(\pi \cdot (1 - \ell)\right) \\ &= \frac{1}{n} \cdot n \left(\pi \cdot (1 - \ell)\right) \\ &= \pi \cdot (1 - \ell)\ . \qquad (17)\end{aligned}$$

Therefore, the naive estimator of $\pi$ is only unbiased whenever $\ell = 0$. Otherwise, the estimator is biased towards 0 the more the persons of the sensitive group lie ($\lim_{\ell \to 1} \mathbb{E}[\pi_{\text{naive}}] = 0$).

**(h)** *Derive expressions for the mean squared error (MSE) of the randomized response estimator and the naive estimator. Analyse the expressions to draw conclusions about when, depending on the relevant parameters $(\pi, a, \ell, n)$, one estimator should be favored over the other.*

We aim to compute the mean squared error of our two estimators of $\pi$.

We use the Bias Variance trade off to decompose the MSE into the bias and variance terms.[2] Therefore, we simplify the MSE into

$$\begin{aligned} \text{MSE}(\hat{\pi}, \pi) &= \mathbb{E}\left[ (\hat{\pi} - \mathbb{E}[\hat{\pi}])^2 \right] + [\mathbb{E}[\hat{\pi}] - \pi]^2 \\ &= \mathbb{V}[\hat{\pi}] + [\mathbb{E}[\hat{\pi}] - \pi]^2 \end{aligned}$$

[2] See https://en.wikipedia.org/wiki/Mean_squared_error#Proof_of_variance_and_bias_relationship.

Therefore, we have Let us first compute $\mathbb{E}[\pi^2_{\text{naive}}]$ and $\mathbb{E}[\pi_{\text{naive}}]^2$.

$$\begin{aligned}
\mathbb{E}[\pi^2_{\text{naive}}] &= \mathbb{E}\left[\left(\frac{\sum_{i=1}^n X_i}{n}\right)^2\right] \\
&= \mathbb{E}\left[\left(\frac{\sum_{i=1}^n X_i}{n}\right)^2\right] \\
&= \mathbb{E}\left[\left(\frac{1}{n}\cdot\sum_{i=1}^n X_i\right)^2\right] \\
&= \mathbb{E}\left[\frac{1}{n^2}\left(\sum_{i=1}^n X_i\right)^2\right] \\
&= \frac{1}{n^2}\mathbb{E}\left[\left(\sum_{i=1}^n X_i\right)^2\right] \\
&= \frac{1}{n^2}\mathbb{E}\left[n\cdot\mathbb{E}[X_i\cdot X_i] + (n^2-n)\cdot\mathbb{E}[X_i\cdot X_j]\right] \\
&= \frac{1}{n^2}\mathbb{E}\left[n\cdot\mathbb{E}[X_i^2] + (n^2-n)\cdot(\mathbb{E}[X_i]+\mathbb{E}[X_j])\right] \\
&= \frac{1}{n^2}\mathbb{E}\left[n\cdot\mathbb{E}[X_i^2] + (n^2-n)\cdot\mathbb{E}[X_i]^2\right] \\
&= \frac{1}{n}\mathbb{E}[X_i^2] + \frac{n^2-n}{n^2}\cdot\mathbb{E}[X_i]^2 \\
&= \frac{1}{n}\mathbb{E}[X_i^2] + \left(1-\frac{1}{n}\right)\cdot\mathbb{E}[X_i]^2 \\
&= \frac{1}{n}\mathbb{E}[X_i^2] + \left(\frac{n-1}{n}\right)\cdot\mathbb{E}[X_i]^2 \\
&= \frac{1}{n}\left[P(X_i=1)\cdot 1^2 + P(X_i=0)\cdot 0^2\right] + \left(\frac{n-1}{n}\right)\cdot\left(P(X_i=1)\cdot 1 + P(X_i=0)\cdot 0\right)^2 \\
&= \frac{1}{n}\left[P(X_i=1)\right] + \left(\frac{n-1}{n}\right)\cdot\left(P(X_i=1)\right)^2 \\
&= \frac{1}{n}\left[\pi\cdot(1-\ell)\right] + \left(\frac{n-1}{n}\right)\cdot\left(\pi\cdot(1-\ell)\right)^2 \\
&= \frac{\pi\cdot(1-\ell)}{n} + \left(\frac{n-1}{n}\right)\cdot\pi^2\cdot(1-\ell)^2 \qquad (18)
\end{aligned}$$

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$$
\begin{aligned}
\mathbb{V}[\pi_{\text{naive}}] &= \mathbb{V}\left[\frac{\sum_{i=1}^{n} X_i}{n}\right] \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{V}[X_i] \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\mathbb{E}[X_i^2] - \mathbb{E}[X_i]^2 \\
&= \frac{1}{n^2}\sum_{i=1}^{n}[P(X_i = 1)] - (P(X_i = 1))^2 \\
&= \frac{1}{n^2}\sum_{i=1}^{n}(\pi \cdot (1-\ell)) - (\pi \cdot (1-\ell))^2 \\
&= \frac{n}{n^2}\left((\pi \cdot (1-\ell)) - (\pi \cdot (1-\ell))^2\right) \\
&= \frac{\pi \cdot (1-\ell) - \pi^2 \cdot (1-\ell)^2}{n}
\end{aligned} \tag{19}
$$

$$
\begin{aligned}
\text{MSE}(\pi_{\text{naive}}, \pi) &= \mathbb{V}[\pi_{\text{naive}}] + [\mathbb{E}[\pi_{\text{naive}}] - \pi]^2 \\
&\overset{(19)}{=} \frac{\pi \cdot (1-\ell) - \pi^2 \cdot (1-\ell)^2}{n} + [\mathbb{E}[\pi_{\text{naive}}] - \pi]^2 \\
&= \frac{\pi \cdot (1-\ell) - \pi^2 \cdot (1-\ell)^2}{n} + \mathbb{E}[\pi_{\text{naive}}]^2 + \pi^2 - 2\pi\mathbb{E}\left[\pi_{\text{naive}}\right] \\
&= \frac{\pi \cdot (1-\ell) - \pi^2 \cdot (1-\ell)^2}{n} + (\pi \cdot (1-\ell))^2 + \pi^2 - 2\pi\pi \cdot (1-\ell) \\
&= \frac{\pi \cdot (1-\ell) - \pi^2 \cdot (1-\ell)^2}{n} + \pi^2 \cdot (1-\ell)^2 + \pi^2 - 2\pi^2 \cdot (1-\ell)
\end{aligned}
$$

# Problem 2

| | | $N = 500$ | $N = 1000$ | $N = 5000$ |
|---|---|---|---|---|
| $\hat{N}$ | | $590.45 \pm 311$ | $1096.62 \pm 286$ | $5152.14 \pm 583$ |
| Error (%) | | 18.09 | 9.66 | 3.04 |
| $\hat{\sigma}_{\hat{N}}$ | | $239.58 \pm 258$ | $284.17 \pm 116$ | $566.36 \pm 98$ |
| C.I. | $\alpha = 0.05$ | $[299.68 \pm 85; 1322.77 \pm 1298]$ | $[687.34 \pm 137; 1842.05 \pm 609]$ | $[4179.35 \pm 423; 6414.78 \pm 804]$ |
| | $\alpha = 0.01$ | $[250.36 \pm 59; 1740.30 \pm 2079]$ | $[601.62 \pm 110; 2185.30 \pm 773]$ | $[3922.46 \pm 383; 6884.52 \pm 890]$ |

Table 1: Result from 1000 MC simulations for estimating $\hat{N}$. For each value, we report on the mean (black) and standard deviation (gray). We omit the digits after the decimal point for the standard deviation for brevity. The Error (%) is the absolute difference between the estimated and the true value of $N$, divided by the true value of $N$. Hence, it measures the relative error of the estimated value to the true value.

**Observation.** We implement the formulas and run the Monte-Carlo simulations for 1000 runs and report the results in Table 1. We first observe that, although the absolute error between the estimated and the true value of $N$ increases with $N$, the relative error decreases when the true $N$ is larger.

Interestingly, the given formula proposed for estimating the variance of $\hat{N}$ is very accurate, as its average value is very close to the computed variance of the Monte-Carlo simulation — in our case, we report on the standard deviation, which is simply the square root of the variance.

Lastly, we computed the averaged confidence intervals across all runs for $\alpha = 0.05$ and $\alpha = 0.01$. Naturally, the confidence intervals when using $\alpha = 0.01$ are larger than when using $\alpha = 0.05$, we notice that the standard deviation of the lower confidence decreases when increasing the size of the Confidence Interval (C.I.), whereas the standard deviation of the upper-confidence increases when decreasing $\alpha$.

**Discussion.**

## Statement on the Useage of Gen. AI

Unless stated otherwise, all code, text, and math derivations have been entirely thought and written by me with no external help — both for gen. AI and wolframalpha anc co.